

Week 5 Assignment: Advanced Java

Task 1 to Task 5 – Concept Explanations

Objective: To understand advanced Java concepts including Packages, Interfaces, Abstract Classes, File Handling, and Multithreading Basics through practical programming.

Task 1: Packages — Organizing Java Code

Explanation: A package in Java is used to organize related classes and interfaces into a separate group. Packages make programs easier to manage, help avoid naming conflicts, and provide better code organization. The **package** keyword is used to create a custom package, while the **import** keyword is used to access a class from another package. In this task, the Student class is placed inside the studentmanagement package and is accessed from another Java class using import.

Task 2: Interface — Payment System

Explanation: An interface in Java defines a common set of methods that implementing classes must provide. It is useful for abstraction and allows different classes to follow the same structure while providing their own implementation. The **implements** keyword is used by a class to implement an interface. In this task, UPIPayment and CardPayment implement the Payment interface, but each class performs payment operations differently.

Task 3: Abstract Classes — Shape Calculator

Explanation: An abstract class is a class declared using the **abstract** keyword. It can contain both abstract methods and normal methods. An abstract method does not have a method body and must be implemented by its child classes. In this task, Shape contains the abstract method calculateArea() and a normal method to display a message. Circle and Rectangle extend Shape and calculate their areas in different ways.

Task 4: File Handling — Student Records

Explanation: File handling in Java is used to create, write, read, and manage data stored in files. Java provides classes such as FileWriter for writing data, FileReader for reading data, and BufferedReader for reading text efficiently. File operations may produce exceptions, so exception handling is required. In this task, student details such as ID, name, course, and marks are written to a text file and then read and displayed in the console.

Task 5: Multithreading Basics — Multiple Tasks

Explanation: Multithreading is a Java feature that allows multiple tasks to execute concurrently within a program. A thread represents an independent path of execution. Threads can be created by extending the Thread class and overriding the run() method. The **start()** method begins thread execution. In this task, one thread prints numbers from 1 to 10 while another thread prints a Java multithreading message. Both threads can execute independently.

Overall Conclusion

These five tasks provide practical understanding of important Advanced Java concepts. Packages improve code organization, interfaces provide a common structure, abstract classes support abstraction, file handling allows programs to work with stored data, and multithreading allows multiple tasks to execute concurrently. Together, these concepts form an important foundation for developing structured and efficient Java applications.